

— NYU SHANGHAI · AI SUMMER PROGRAM 2026

Second Life AI

Six-Class Waste Sorting, Two Ways

A compact ConvNeXt V2 under the 30M-parameter cap, trained beside a Swin-B transformer three times its size. Both reach the high-90s on group-disjoint splits — and a live demo lets anyone photograph their own object and watch the two models answer.

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OURS · CONVNEXT V2 TINY

98.30%

validation · 27.9M params · under the cap

TRANSFORMER · SWIN-B

97.61%

held-out test · 86.8M params

Different corpora and splits — read as two strong models, not a head-to-head. The shared test is the live demo.

01 PROBLEM & DATA

Waste, as it is photographed

Can a model sort the photo someone actually takes at a bin — varied cameras, lighting and backgrounds — into six recyclable materials?

SIX CANONICAL CLASSES

■ Cardboard

■ Metal

■ Plastic

■ Glass

■ Paper

■ Trash

POOLED PUBLIC DATA

One dataset shot against one backdrop teaches a model one studio. We train on broad public waste imagery instead — a labelled waste-management corpus plus **RealWaste** (UCI), spanning many cameras, angles and real backgrounds.

Near-identical copies are removed by **perceptual hash** before anything is split, so the same object cannot sit on two sides of the line.

PARAMETER BUDGET · 30M CAP

ConvNeXt V2 Tiny · ours27.9M

Swin-B · transformer86.8M

Our model spends its budget on a 384 px input and strong pretraining — and still clears the cap with room to spare.

02 EVALUATION DISCIPLINE

Split by object, not by image

Every reported number comes from a split where whole near-duplicate groups — not individual photos — are held out, so a score reflects the material, not a memorised object.

HOW THE NUMBERS STAY HONEST

✓ Group-disjoint **StratifiedGroupKFold** — no object crosses the split

✓ Augmentation applied **only after** splitting, to training images

✓ Fixed seed 42; committed split identities

✓ Every run logged with a configuration hash

03 THE TWO MODELS

Small under a cap, and large

	OURS	TRANSFORMER
Backbone	ConvNeXt V2 T	Swin-B
Parameters	27.9M	86.8M
Under 30M cap	Yes	No
Input	384 px	224 px
Pretraining	FCMAE→IN-22k	IN-1k
Accuracy	98.3% val	97.6% test

Our model reaches the transformer's league at **a third of the parameters** — the budget goes to data and input resolution, not size.

TRAINING RECIPE · OURS

two-phase fine-tune · AdamW · cosine · head 2e-4 / backbone 3e-5 · label smoothing 0.05 · Mixup 0.2 + CutMix 0.5 · EMA 0.999 · AMP · seed 42

A 2-epoch head warmup unfreezes into 58 discriminative-rate epochs; best checkpoint at epoch 37.

AUGMENTATION · TRAIN ONLY

RandomResizedCrop 384

Horizontal flip

TrivialAugmentWide

RandomErasing 0.20

Evaluation is deterministic: resize to 438, centre-crop 384. Mixup and CutMix switch off for the final three epochs so the model finishes on clean labels.

60EPOCHS

37BEST @

64EFF. BATCH

04 RESULTS

Strong on every class

PER-CLASS RECALL · SWIN-B · HELD-OUT TEST + TTA	RECALL
Metal	98.6%
Cardboard	98.1%
Glass	97.3%
Paper	96.5%
Plastic	95.4%
Trash	94.1%

No class drops below 94%. **Trash** — the smallest, most varied class — remains the hardest, as expected.

KEY RESULT

A 27.9M-parameter model, under the competition cap, reaches **98.3% validation** — level with a transformer three times larger. Discipline and data, not scale.

05 DEPLOYMENT

A model that knows its limits

- A live site runs **both models on the same photo**, side by side — an offline Flask app, no network needed.

A six-slot classifier must label **even a photo of a cat**. Our feature-space guard flags it as out-of-domain (distance 0.78 > 0.70) where the transformer answers “paper” with confidence.

Photograph your own object — that fresh, never-seen image is the fair test both models share.

OPEN-SET GUARD · A PHOTO OF A CAT

Transformer“paper”, confident

Oursflagged · dist 0.78 > 0.70

A feature-space nearest-neighbour distance catches inputs that belong to none of the six classes — the kind of refusal that matters at a real bin.

SERVING

✓ Offline Flask app — weights are local, no downloads

✓ Both models load once; toggled per request

✓ Single deterministic view each, own eval transform

✓ Vended React front end · bilingual zh / en

TRY IT LIVE · TWO-MODEL DEMO

Point your camera at anything in the room. Toggle ConvNeXt and Swin-B on the same shot, and watch where a big model is confidently wrong and a careful one holds back.

bilingual · zh / en161-33-76-49.sslip.io

Sources: RealWaste (UCI, 2023) · a labelled public waste-management corpus · garbage_classification. Accuracies are each model's own held-out split — ConvNeXt V2 validation, Swin-B test — so they are read as two strong models, not a direct ranking; the fair shared test is fresh, never-seen photos in the live demo. Values trace to best_convnextv2_metadata.json and swin_b_results.json.